

Aria Assembly Documentation

June 14, 2026

This document provides the standard operating procedure and technical specifications for the assembly of the pneumatic belt tracking system.

Assembly Procedure

Assembly: The pneumatic tubing is cut into two 40cm segments and connected to a T-joint. The open extremities are sealed by folding the ends and securing them with cable ties. The tube assembly is then fixed to the upper edge of the elastic belt, oriented so that the T-joint rests over the sternum when the belt is fastened, pointing downwards, with the closure mechanism at the back. The tube is initially secured using medical tape, which is then reinforced by sewing it into the belt. These stitches are applied carefully to avoid piercing the tubing, ensuring the structural stability of the system while maintaining its integrity.

The differential pressure sensor is connected with the perpendicular port of the T-joint using a shorter segment of smaller tubing. To ensure the integrity of the circuit, the connection is reinforced with an adhesive to establish a hermetic seal. Finally, the sensor housing is glued directly to the outer surface of the belt, ensuring it remains stable and protected from mechanical stress caused by the user's torso movements.

System Specifications

Below is a summary of the required structural components and the wiring diagram for the Arduino microcontroller.

Table 1: Bill of Materials.

Component	Specification	Supplier	Qty	Cost (€)
Arduino UNO R4 WiFi	Renesas RA4M1, WiFi, USB-C	Arduino	1	30.50
MPU6050	6-axis IMU, I ² C, 3.3V/5V	Arceli	1	4.70
MPXV7002DP	Differential Pressure, I ² C	Culnfun	1	34.79
Silicone tube	ID 4mm, OD 6mm, 80cm	Stubotar	1	4.50
Silicone tube	ID 2mm, OD 4mm, 5cm	—	1	0.10
T-joint	OD 6mm	Gardena	1	1.98
Vibration motor	11k $\pm$ 2.5k RPM, 3V, 80mA max	Garosa	2	4.50
Elastic band	16.5 cm, polyester	Tempo di saldi	1	11.00
9V battery	2000mAh, 9V	Energizer	1	2.60
Misc.	Resistors, transistors, connectors	—	1	—
Total				94.67

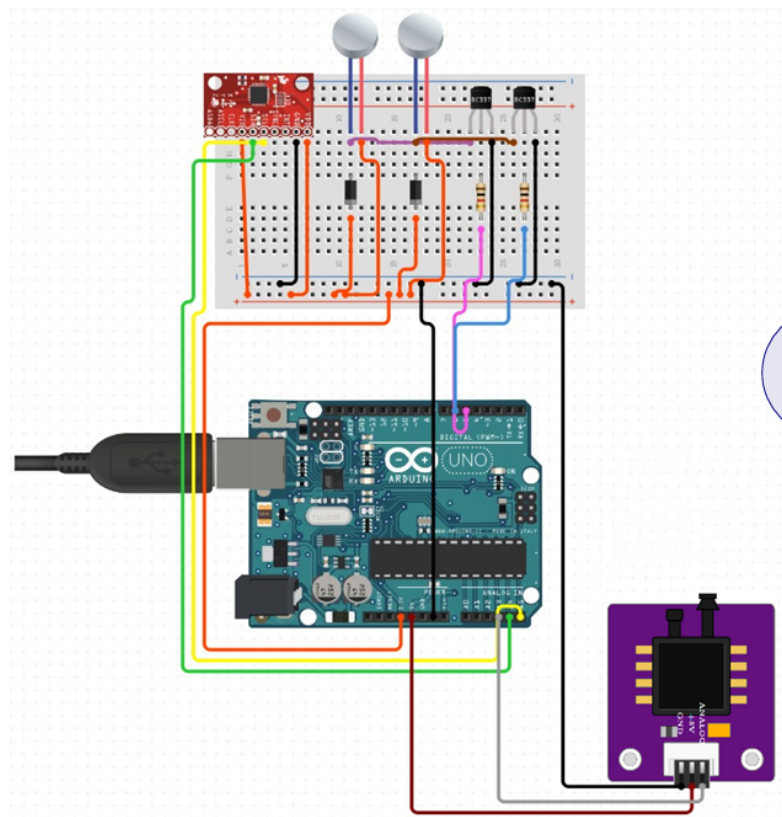


Figure 1: Schematic diagram of the physical system layout.